Lightning Intro to Tensorflow/Keras

What’s a Tensor?

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Tensor of rank 0 (Scalar)

42

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Tensor of rank 0 (Scalar)

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Tensor of rank 1 (aka Vector)

(42, 23.4, 11.2)

What's a Tensor?

Tensor of rank 0 (Scalar)

42

Tensor of rank 2 (aka Matrix)

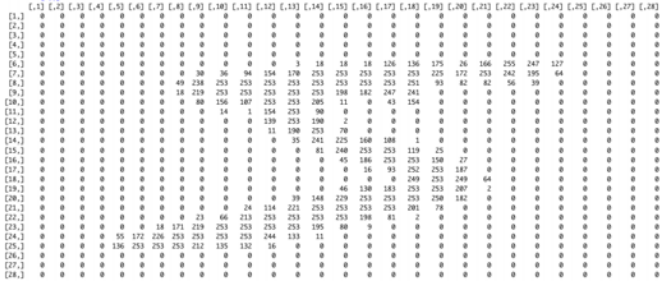


Image credit: fast.ai

Tensor of rank 1 (aka Vector)

(42, 23.4, 11.2)

What's a Tensor?

Tensor of rank 0 (Scalar)

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Tensor of rank 1 (aka Vector)

(42, 23.4, 11.2)

Tensor of rank 2 (aka Matrix)

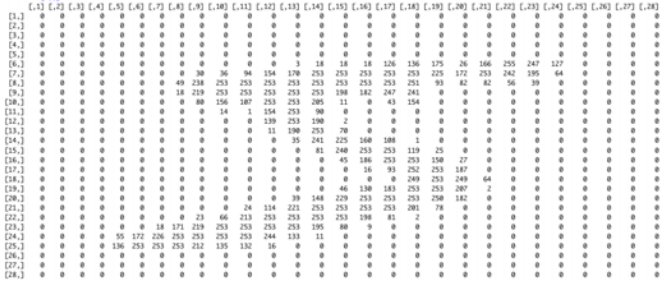
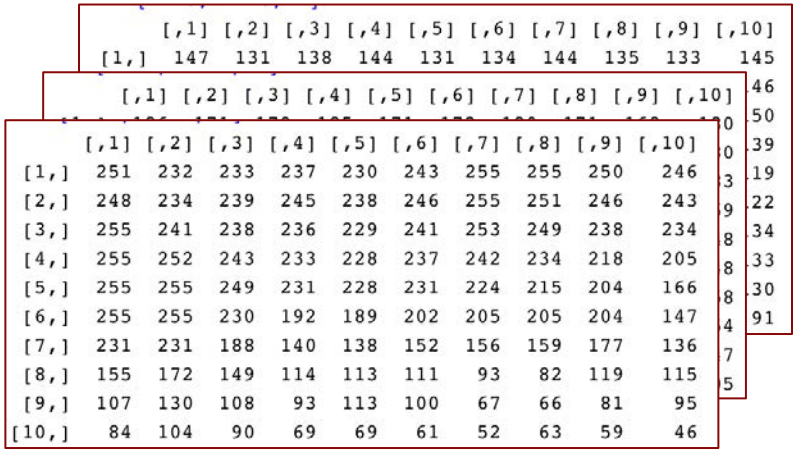


Image credit: fast.ai

Tensor of rank 3 (aka "cube")



	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	251	232	233	237	230	243	255	255	250	246
[2,]	248	234	239	245	238	246	255	251	246	243
[3,]	255	241	238	236	229	241	253	249	238	234
[4,]	255	252	243	233	228	237	242	234	218	205
[5,]	255	255	249	231	228	231	224	215	204	166
[6,]	255	255	230	192	189	202	205	205	204	147
[7,]	231	231	188	140	138	152	156	159	177	136
[8,]	155	172	149	114	113	111	93	82	119	115
[9,]	107	130	108	93	113	100	67	66	81	95
[10,]	84	104	90	69	69	61	52	63	59	46

Can you give an example of a rank-4 tensor?

What's a Tensor?

See Chapter 2.2 of text for more detail

Tensorflow

Tensorflow (TF) is a library that provides

- Automatic calculation of the gradient of (complicated) loss functions

$$\nabla Loss(w) = \left[\frac{\partial Loss}{\partial w_1}, \frac{\partial Loss}{\partial w_2}, \ldots, \frac{\partial Loss}{\partial wn}\right]$$

Tensorflow

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- Library of state-of-the-art optimizers

Tensorflow

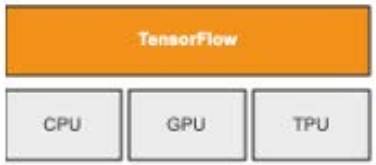
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Tensorflow

Tensorflow (TF) is a library that provides

- Automatic calculation of the gradient of (complicated) loss functions
- Library of state-of-the-art optimizers
- Automatic distribution of computational load across servers
- Automatic adaptation of code to work on parallel hardware (GPUs and TPUs)



Keras “sits on top of” Tensorflow …

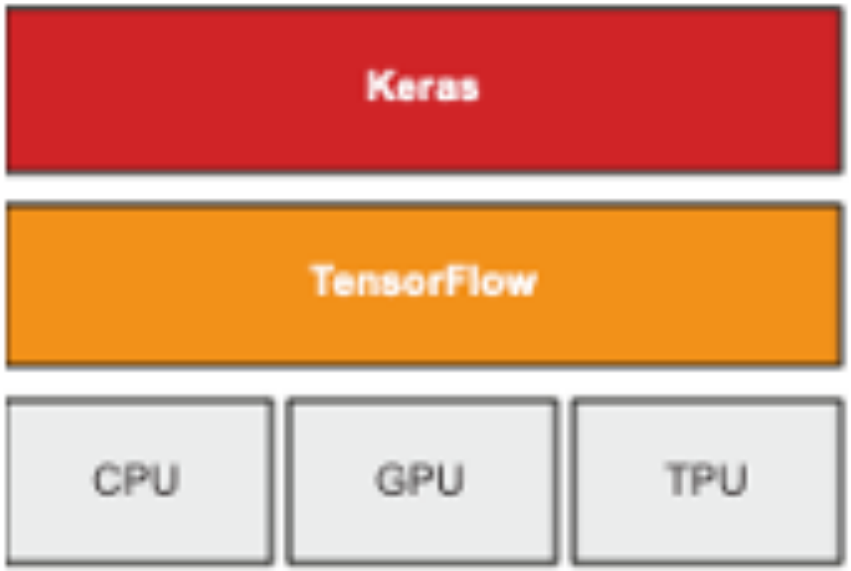


Image: Page 70 of textbook

… and provides “convenience” features

- Pre-defined layers
- Incredibly flexible ways to specify network architectures
- Easy ways to preprocess data
- Easy ways to train models and report metrics
- Pre-trained models you can download and customize

Keras APIs

- There are three broad ways to build DL models with Keras
  - Sequential
  - Functional API
  - Subclassing

- We will almost exclusively use the Functional API. The model we built for heart disease prediction is an example.

- Please read 7.2.2 of the textbook to understand in detail how the Keras Functional API works